Full Molecular Hamiltonian

From Classical to Quantum: the exact non-relativistic quantization

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Roadmap: four stages

We build the exact non-relativistic molecular Hamiltonian in four stages:

1. **Classical** — lab-frame H; one point transformation separates the centre of mass *and* produces the mass polarization.
2. **Body-Fixed** — co-rotating frame + Lagrangian $\Rightarrow$ classical molecular H_{int}.
3. **Quantize** — Podolsky rule from coordinate derivatives alone: metric $\to$ measure $\to$ inverse $\to$ Watson pseudopotential. No angular-momentum algebra.
4. **Quantum** $\hat{H}$ — the exact non-relativistic molecular Hamiltonian.

Outline

Motivation

Why is the molecular Hamiltonian non-trivial?

The lab Hamiltonian couples *all* particles through the Coulomb potential:

$$H = \sum_j \frac{\mathbf{P}_j^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V.$$

To do molecular physics we want to separate

$$\underbrace{\text{translation}}_{\text{COM}} \Big| \underbrace{\text{rotation}}_{\text{Euler angles } \Omega} \Big| \underbrace{\text{vibration}}_{\text{normal coords } Q} \Big| \underbrace{\text{electrons}}_{\bar{\mathbf{r}}_i}.$$

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The catch

Rotation + vibration are *curvilinear* coordinates. Naively quantizing $p \to -i\hbar\,\partial$ is **not** Hermitian there. Getting it right forces the Podolsky construction.

What we will not approximate

We keep everything exact until the very end:

- **No** Born–Oppenheimer / clamped nuclei.
- **No** neglect of *mass polarization*.
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The *only* ingredients are:

1. non-relativistic dynamics;
2. the Eckart frame (a *convention*, not an approximation);
3. parametrizing the nuclear shape by $3N_{\mathrm{nucl}} - 6$ normal coordinates (exact).

The endpoint is the exact quantum $\hat{H}_{\mathrm{int}}$ — the rigorous starting point

Classical Cartesian Hamiltonian

The Lagrangian in the lab frame

N_{nucl} nuclei $(M_j, Z_j e, \mathbf{R}_j)$ and N_{elec} electrons $(m_e, -e, \mathbf{r}_i)$, all in the space-fixed frame; SI units. The Lagrangian is kinetic minus potential:

$$\mathcal{L} = \sum_j \tfrac{1}{2} M_j \dot{\mathbf{R}}_j^2 + \sum_i \tfrac{1}{2} m_e \dot{\mathbf{r}}_i^2 - V(\mathbf{R}_j, \mathbf{r}_i).$$

Differentiate $\mathcal{L}$ with respect to each velocity to get the canonical momenta:

$$\boxed{\mathbf{P}_j \equiv \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{R}}_j} = M_j \dot{\mathbf{R}}_j, \qquad \mathbf{p}_i \equiv \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}_i} = m_e \dot{\mathbf{r}}_i.}$$

Canonical brackets $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk}\delta_{\alpha\beta}$, $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij}\delta_{\alpha\beta}$.

- Kinetic energy is diagonal and quadratic in Cartesian velocities: *canonical momentum* $=$ *mass* $\times$ *velocity*, no surprises.
- This is the *only* stage of the whole paper where canonical and mechanical momentum coincide.

Legendre transform to the Hamiltonian

The three-move recipe (we reuse it, less trivially, in a moment):

1. **Define** $p_q \equiv \partial\mathcal{L}/\partial\dot{q}$ for every coordinate.
2. **Invert** for every velocity as a function of the momenta.
3. **Assemble** $H \equiv \sum_q p_q \dot{q} - \mathcal{L}$ and substitute.

Here step 2 is immediate, $\dot{\mathbf{R}}_j = \mathbf{P}_j/M_j$, $\dot{\mathbf{r}}_i = \mathbf{p}_i/m_e$; step 3 gives

$$H = \underbrace{\sum_j \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_i \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \tag{1.1}$$

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{1}{2}\sum_{i\neq j}\frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j}\frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{r_{ij}}\right). \tag{1.2}$$

Everything that follows is an exact rewriting of (1.1)–(1.2).

Mass-centered coordinates

Two centers of mass, two roles

The translational coordinate and the internal reference point need not be the same. Define

$$M_N = \sum_j M_j, \qquad M_{\text{tot}} = M_N + N_{\text{elec}} m_e,$$

$$\mathbf{R}_{\text{ncm}} = \frac{1}{M_N}\sum_j M_j \mathbf{R}_j, \qquad \mathbf{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}}\Big(\sum_j M_j \mathbf{R}_j + m_e \sum_i \mathbf{r}_i\Big), \tag{A.2, A.3}$$

and refer *every* internal coordinate — nuclear and electronic — to the *nuclear* CoM:

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{ncm}}, \qquad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{ncm}}, \qquad \textstyle\sum_j M_j \mathbf{R}'_j = 0. \tag{A.4}$$

The mixed choice

Translation = total CoM ⇒ correct total mass, and (we will see) the COM–internal cross terms cancel identically.

Internals referred to the nuclear CoM ⇒ Born–Oppenheimer-friendly coordinates, V form-invariant, and the mass polarization appears *by itself*.

Inverting: every particle rides a shared drift

Solve (A.3) for the nuclear CoM, then invert (A.4) for the lab positions:

$$\mathbf{R}_{\mathrm{ncm}} = \mathbf{R}_{\mathrm{cm}} - \frac{m_e}{M_{\mathrm{tot}}} \sum_l \mathbf{r}'_l, \tag{A.5}$$

$$\mathbf{R}_j = \mathbf{R}_{\mathrm{cm}} + \mathbf{R}'_j - \frac{m_e}{M_{\mathrm{tot}}} \sum_l \mathbf{r}'_l, \qquad \mathbf{r}_i = \mathbf{R}_{\mathrm{cm}} + \mathbf{r}'_i - \frac{m_e}{M_{\mathrm{tot}}} \sum_l \mathbf{r}'_l. \tag{A.6}$$

Both families ride on the *same* drift — the nuclear-CoM velocity:

$$\boxed{\mathbf{D} \equiv \dot{\mathbf{R}}_{\mathrm{ncm}} = \dot{\mathbf{R}}_{\mathrm{cm}} - \frac{m_e}{M_{\mathrm{tot}}} \sum_l \dot{\mathbf{r}}'_l, \qquad \dot{\mathbf{R}}_j = \mathbf{D} + \dot{\mathbf{R}}'_j, \qquad \dot{\mathbf{r}}_i = \mathbf{D} + \dot{\mathbf{r}}'_i.} \tag{A.7}$$

Every lab velocity = internal velocity + one shared drift $\mathbf{D}$ *— passengers on a train.*

Kinetic energy in velocity form: the mass polarization appears

Substitute (A.7) into $T = \sum_j \frac{1}{2} M_j \dot{\mathbf{R}}_j^2 + \sum_i \frac{1}{2} m_e \dot{\mathbf{r}}_i^2$. The nuclear cross term dies by $\sum_j M_j \dot{\mathbf{R}}'_j = 0$; the electronic one does *not*:

$$T_N = \tfrac{1}{2} M_N \mathbf{D}^2 + \tfrac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2, \qquad T_e = \tfrac{1}{2} N_{\mathrm{elec}} m_e \mathbf{D}^2 + m_e \mathbf{D} \cdot \sum_i \dot{\mathbf{r}}'_i + \tfrac{1}{2} m_e \sum_i \dot{\mathbf{r}}_i'^2. \quad \text{(A.8, A.9)}$$

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Expand $\mathbf{D}$ in the two $\mathbf{D}$-dependent pieces (A.11a,b): the $\dot{\mathbf{R}}_{\mathrm{cm}} \cdot \sum \dot{\mathbf{r}}'$ cross terms cancel identically, and the quadratic pieces combine to one *negative* remainder:

$$\boxed{T = \tfrac{1}{2} M_{\mathrm{tot}} \dot{\mathbf{R}}_{\mathrm{cm}}^2 + \tfrac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2 + \tfrac{1}{2} m_e \sum_i \dot{\mathbf{r}}_i'^2 - \frac{m_e^2}{2 M_{\mathrm{tot}}} \Big(\sum_i \dot{\mathbf{r}}'_i \Big)^2.} \qquad \text{(A.12)}$$

The CoM decouples with the *correct* mass M_{tot}; the electrons carry a negative collective correction — the mass polarization in disguise.

Canonical momenta: $\partial\mathcal{L}/\partial\dot{q}$, and the tricky one

With $\mathcal{L} = T - V$ (V velocity-independent), differentiate the boxed T:

$$\mathbf{P}_{\mathrm{cm}} = \frac{\partial\mathcal{L}}{\partial\dot{\mathbf{R}}_{\mathrm{cm}}} = M_{\mathrm{tot}}\dot{\mathbf{R}}_{\mathrm{cm}}, \qquad \mathbf{P}'_j = \frac{\partial\mathcal{L}}{\partial\dot{\mathbf{R}}'_j} = M_j\dot{\mathbf{R}}'_j. \tag{A.13, A.17}$$

The electrons take care: *two* terms of T involve $\dot{\mathbf{r}}'_i$ (its own kinetic term, *and* the mass-polarization sum), so

$$\boxed{\mathbf{p}'_i \equiv \frac{\partial\mathcal{L}}{\partial\dot{\mathbf{r}}'_i} = m_e\dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\mathrm{tot}}}\sum_l \dot{\mathbf{r}}'_l \neq m_e\dot{\mathbf{r}}'_i.} \tag{A.14}$$

Canonical $\neq$ mechanical: moving *one* electron drags the whole molecule's CoM, and $\mathbf{r}'_i$ is measured from the (recoiling) nuclear CoM — every electron's momentum carries a shared correction.

Inverting the electron relation

Sum (A.14) over all electrons; with $M_{\mathrm{tot}} - N_{\mathrm{elec}}\, m_e = M_N$,

$$\sum_i \mathbf{p}'_i = m_e\left(1 - \frac{N_{\mathrm{elec}}\, m_e}{M_{\mathrm{tot}}}\right)\sum_l \dot{\mathbf{r}}'_l = \frac{m_e M_N}{M_{\mathrm{tot}}}\sum_l \dot{\mathbf{r}}'_l. \tag{A.15}$$

Solve for $\dot{\mathbf{r}}'_i$:

$$\boxed{\dot{\mathbf{r}}'_i = \frac{\mathbf{p}'_i}{m_e} + \frac{1}{M_N}\sum_l \mathbf{p}'_l.} \tag{A.16}$$

Each electron's velocity = its own momentum/m_e + one shared recoil — the momentum-space mirror of the shared drift $\mathbf{D}$.

Legendre transform: kinetic energy in momentum form

CoM and nuclear terms are immediate: $\frac{1}{2}M_{\rm tot}\dot{\mathbf{R}}_{\rm cm}^2 = \mathbf{P}_{\rm cm}^2/2M_{\rm tot}$, $\frac{1}{2}M_j\dot{\mathbf{R}}_j'^2 = \mathbf{P}_j'^2/2M_j$.
Substituting (A.16) into the electron sector (A.19, A.20) and collecting the $\left(\sum_l \mathbf{p}_l'\right)^2$ coefficients,

$$\frac{1}{M_N} + \frac{N_{\rm elec}m_e - M_{\rm tot}}{2M_N^2} = \frac{1}{M_N} - \frac{M_N}{2M_N^2} = +\frac{1}{2M_N}. \tag{A.21}$$

The negative velocity-form correction has become the positive mass polarization:

$$\boxed{T = \frac{\mathbf{P}_{\rm cm}^2}{2M_{\rm tot}} + \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N}\left(\sum_i \mathbf{p}_i'\right)^2.} \tag{A.22}$$

The Hamiltonian — and why the sign flips

Both internal families share the reference point $\mathbf{R}_{\mathrm{ncm}}$, so $V(\mathbf{R}'_j, \mathbf{r}'_i)$ is unchanged in form. Add it to (A.22):

$$\boxed{H = \frac{\mathbf{P}_{\mathrm{cm}}^{\,2}}{2M_{\mathrm{tot}}} + \underbrace{\sum_j \frac{\mathbf{P}_j'^{\,2}}{2M_j} + \sum_i \frac{\mathbf{p}_i'^{\,2}}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \mathbf{p}'_i\Big)^2 + V(\mathbf{R}'_j, \mathbf{r}'_i)}_{H_{\mathrm{int}}\ (2.13)}\,.} \qquad (2.12 \equiv \text{A.23})$$

Why the sign flips

The electron block of T is the quadratic form $\frac{1}{2}\dot{\mathbf{r}}'^{\mathsf{T}}\mathbf{A}\dot{\mathbf{r}}'$ with $\mathbf{A} = m_e(\mathbb{1} - \frac{m_e}{M_{\mathrm{tot}}}\mathbf{J})$ ($\mathbf{J}$ all-ones); the Legendre transform replaces $\mathbf{A} \to \mathbf{A}^{-1} = \frac{1}{m_e}(\mathbb{1} + \frac{m_e}{M_N}\mathbf{J})$ (A.25, A.26) — same physics, opposite side of the transform.

Exact for any $\mathbf{P}_{\mathrm{cm}}$: no rest frame, no boost, no cross-coupling — agrees with the paper's generating-function route (Sec. 2). *H_{int} is what we carry into the body-fixed frame.*

Body-fixed frame and the Lagrangian

Body-fixed frame: rotate with the nuclei

Introduce a rotation $\mathbf{S}(\Omega)$ (Euler angles $\Omega = \phi, \theta, \chi$) and body-fixed (BF) coordinates:

$$\mathbf{R}'_j = \mathbf{S}(\Omega)\,\bar{\mathbf{R}}_j(Q), \qquad \mathbf{r}'_i = \mathbf{S}(\Omega)\,\bar{\mathbf{r}}_i. \tag{3.1}$$

The *Eckart* conditions fix $\mathbf{S}$ from the *nuclei*; the nuclear shape is rectilinear mass-weighted normal coordinates Q_b:

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\mathrm{eq}} + \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb}\,Q_b, \qquad \sum_j \mathbf{l}_{ja}\cdot\mathbf{l}_{jb} = \delta_{ab}. \tag{3.1a}$$

The BF angular velocity $\boldsymbol{\omega}$ is defined by

$$(\mathbf{S}^T\dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma}\,\omega_\gamma. \tag{3.2}$$

The velocity-form Lagrangian — everything, electrons included

Take the internal energy of Sec. 2 in *velocity* form (before the Legendre transform, eq. A.12) and body-fix it: substitute the co-rotating velocities (3.3) $\dot{\mathbf{R}}'_j = \mathbf{S}(\omega\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j)$, $\dot{\mathbf{r}}'_i = \mathbf{S}(\omega\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i)$, using $|\mathbf{S}\mathbf{v}|^2 = |\mathbf{v}|^2$. Every particle stays in velocity form:

$$\boxed{\mathcal{L} = \tfrac{1}{2}M_{\mathrm{tot}}|\dot{\mathbf{R}}_{\mathrm{cm}}|^2 + \tfrac{1}{2}\sum_j M_j|\omega\times\bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 + \tfrac{m_e}{2}\sum_i|\omega\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i|^2 - \frac{m_e^2}{2M_{\mathrm{tot}}}|\omega\times\bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}|^2 - V} \tag{3.7$'$}$$

with $\bar{\mathbf{r}} \equiv \sum_i \bar{\mathbf{r}}_i$. The potential is now manifestly $\mathbf{S}$-independent (3.8). We carry the electrons in velocity form all the way — $\mathbf{I}_e$, $\mathbf{I}_X$ and all.

The vibrational term: $\dot{\bar{\mathbf{R}}}_j \to \dot{Q}_b$

Expanding each square $|\boldsymbol{\omega}\times\mathbf{a}+\dot{\mathbf{a}}|^2$ leaves a pure-velocity (vibrational) piece $|\dot{\mathbf{a}}|^2$. For the nuclei it collapses cleanly: the mode vectors are *constant* $[\partial\bar{\mathbf{R}}_j/\partial Q_b = \mathbf{l}_{jb}/\sqrt{M_j}$ from (3.1a)], so

$$\dot{\bar{\mathbf{R}}}_j = \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb}\,\dot{Q}_b.$$

The M_j cancels and the orthonormality $\sum_j \mathbf{l}_{jb}\cdot\mathbf{l}_{jc} = \delta_{bc}$ (3.1a) collapses the double sum:

$$\boxed{T^N_{\mathrm{vib}} \equiv \tfrac{1}{2}\sum_j M_j|\dot{\bar{\mathbf{R}}}_j|^2 = \tfrac{1}{2}\sum_{b,c}\Big(\underbrace{\textstyle\sum_j \mathbf{l}_{jb}\cdot\mathbf{l}_{jc}}_{=\delta_{bc}}\Big)\dot{Q}_b\dot{Q}_c = \tfrac{1}{2}\sum_b \dot{Q}_b^2.}$$

The vibrational metric is the identity — no Wilson G-matrix. The electron kinetic and mass-polarization pieces stay in velocity form; every symbol is collected on the next slide.

Expand every squared term ⇒ the full inertia

The vector identity (any $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega}$)

$$\frac{m}{2}|\boldsymbol{\omega}\times\mathbf{a}+\dot{\mathbf{a}}|^2 = \underbrace{\frac{m}{2}\boldsymbol{\omega}^T(a^2\mathbb{1}-\mathbf{a}\mathbf{a}^T)\boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m\,\boldsymbol{\omega}\cdot(\mathbf{a}\times\dot{\mathbf{a}})}_{\text{Coriolis}} + \underbrace{\frac{m}{2}|\dot{\mathbf{a}}|^2}_{\text{vibrational}} \tag{3.9c}$$

applied to *nuclei, electrons, and the mass-polarization term* gives

$$\boxed{\mathcal{L} = \tfrac{1}{2}M_{\rm tot}|\dot{\mathbf{R}}_{\rm cm}|^2 + \tfrac{1}{2}\boldsymbol{\omega}^T\underbrace{(\mathbf{I}_N+\mathbf{I}_e-\mathbf{I}_X)}_{\text{full inertia }\mathbf{I}}\boldsymbol{\omega} + \boldsymbol{\omega}\cdot(\mathbf{L}_N^{\rm vib}+\mathbf{L}_e-\mathbf{L}_X) + \tfrac{1}{2}\sum_b \dot{Q}_b^2 + T_e^{\rm BF} - T_X^{\rm mp} - V} \tag{3.13'}$$

Definitions. *Inertia* (3.9,3.10): $\mathbf{I}_N = \sum_j M_j(\bar{R}_j^2\mathbb{1}-\bar{\mathbf{R}}_j\bar{\mathbf{R}}_j^T)$, $\mathbf{I}_e = m_e\sum_i(\bar{r}_i^2\mathbb{1}-\bar{\mathbf{r}}_i\bar{\mathbf{r}}_i^T)$, $\mathbf{I}_X = \frac{m_e^2}{M_{\rm tot}}(\bar{r}^2\mathbb{1}-\bar{\mathbf{r}}\bar{\mathbf{r}}^T)$. *Angular momenta* (3.11): $\mathbf{L}_N^{\rm vib} = \sum_j M_j\bar{\mathbf{R}}_j\times\dot{\bar{\mathbf{R}}}_j$, $\mathbf{L}_e = m_e\sum_i\bar{\mathbf{r}}_i\times\dot{\bar{\mathbf{r}}}_i$, $\mathbf{L}_X = \frac{m_e^2}{M_{\rm tot}}\bar{\mathbf{r}}\times\dot{\bar{\mathbf{r}}}$. *Kinetic pieces* (3.12): $T_e^{\rm BF} = \frac{m_e}{2}\sum_i|\dot{\bar{\mathbf{r}}}_i|^2$, $T_X^{\rm mp} = \frac{m_e^2}{2M_{\rm tot}}|\dot{\bar{\mathbf{r}}}|^2$ (with $\frac{1}{2}\sum_b\dot{Q}_b^2 = T_{\rm vib}^N$ from the previous slide).

Total, nuclear, and canonical electronic angular momenta

Read the momenta off $\mathcal{L}$ (3.13′):

$$\mathbf{J} \equiv \frac{\partial \mathcal{L}}{\partial \boldsymbol{\omega}} = \underbrace{(\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}} + \mathbf{L}_e - \mathbf{L}_X}_{\text{total ang. mom.}}, \qquad \mathbf{J}_N \equiv \mathbf{I}_N\boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}} \quad \text{(nuclear)}. \quad (3.16,\ 3.15)$$

The canonical electronic angular momentum is built from the electron pair $(\bar{\mathbf{r}}_i, \bar{\mathbf{p}}_i)$:

$$\mathbf{L}_{\mathrm{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i, \qquad \bar{\mathbf{p}}_i \equiv \frac{\partial \mathcal{L}}{\partial \dot{\bar{\mathbf{r}}}_i} = m_e(\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i) - \frac{m_e^2}{M_{\mathrm{tot}}}(\boldsymbol{\omega} \times \bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}). \qquad (3.14)$$

Vibrational momentum, with the Coriolis vectors $\boldsymbol{\sigma}_b$:

$$P_b = \frac{\partial \mathcal{L}}{\partial \dot{Q}_b} = \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b + \dot{Q}_b, \qquad \boldsymbol{\sigma}_b = \sum_a \zeta_{ab} Q_a, \quad \zeta_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}. \qquad (3.17\text{a,b})$$

Proof 1: $\mathbf{L}_{\mathrm{elec}}$ is the electron + mass-polarization ang. mom.

Substitute $\bar{\mathbf{p}}_i$ (3.14) into $\mathbf{L}_{\mathrm{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ and use BAC–CAB on the two $\boldsymbol{\omega}$-terms,

$$\mathbf{a} \times (\boldsymbol{\omega} \times \mathbf{a}) = (a^2 \mathbb{1} - \mathbf{a}\mathbf{a}^T)\boldsymbol{\omega} :$$

$$\boxed{\mathbf{L}_{\mathrm{elec}} = (\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X).} \tag{3.18}$$

What this says

The canonical electronic angular momentum is *exactly* the electron and mass-polarization inertia/Coriolis pieces of $\mathcal{L}$ (3.13′). Therefore the total splits cleanly,

$$\mathbf{J} = \underbrace{\mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}}}_{\mathbf{J}_N} + \underbrace{(\mathbf{I}_e - \mathbf{I}_X)\boldsymbol{\omega} + (\mathbf{L}_e - \mathbf{L}_X)}_{\mathbf{L}_{\mathrm{elec}}} = \mathbf{J}_N + \mathbf{L}_{\mathrm{elec}}. \tag{3.16}$$

Proof 2: the electron sector recovers H_e

Write the body-frame electron velocity $\mathbf{w}_i \equiv \boldsymbol{\omega}\times\bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i$. The electron + mass-polarization part of $\mathcal{L}$ (3.7′) is

$$\mathcal{L}_e^{\mathrm{vel}} = \frac{m_e}{2}\sum_i |\mathbf{w}_i|^2 - \frac{m_e^2}{2M_{\mathrm{tot}}}\Big|\sum_i \mathbf{w}_i\Big|^2, \qquad \bar{\mathbf{p}}_i = m_e\mathbf{w}_i - \frac{m_e^2}{M_{\mathrm{tot}}}\sum_l \mathbf{w}_l,$$

identical in structure to the Sec. 2 velocity form (A.12) with $\dot{\mathbf{r}}_i' \to \mathbf{w}_i$. The same mass-matrix inversion $\mathbf{A} \to \mathbf{A}^{-1}$ (A.25–A.26) gives

$$\boxed{\sum_i \bar{\mathbf{p}}_i\cdot\mathbf{w}_i - \mathcal{L}_e^{\mathrm{vel}} = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2 \equiv H_e.}$$

Why ω does not spoil it

In $\sum p\dot{q}$, use $\dot{\bar{\mathbf{r}}}_i = \mathbf{w}_i - \boldsymbol{\omega}\times\bar{\mathbf{r}}_i$: $\sum_i \bar{\mathbf{p}}_i\cdot\dot{\bar{\mathbf{r}}}_i = \sum_i \bar{\mathbf{p}}_i\cdot\mathbf{w}_i - \boldsymbol{\omega}\cdot\mathbf{L}_{\mathrm{elec}}$. The $-\boldsymbol{\omega}\cdot\mathbf{L}_{\mathrm{elec}}$ cancels the electronic part of $\mathbf{J}\cdot\boldsymbol{\omega}$, leaving $\mathbf{J}_N\cdot\boldsymbol{\omega}$ for rotation and the clean H_e for electrons.

The key identity

Combining (3.16) and (3.18), the electronic and mass-polarization inertias cancel between $\mathbf{J}$ and $\mathbf{L}_{\rm elec}$:

$$\boxed{\mathbf{J} - \mathbf{L}_{\rm elec} = \mathbf{I}_N(Q)\,\boldsymbol{\omega} + \mathbf{L}_N^{\rm vib}.} \tag{3.19}$$

The algebraic miracle

The full inertia $\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X$ appeared in $\mathcal{L}$, but $\mathbf{I}_e, \mathbf{I}_X$ drop out of the right-hand side: the surviving rotational kernel is the *nuclear* inertia $\mathbf{I}_N$ **alone**. All electronic + mass-polarization content sits in the kinematic shift $\mathbf{J} \to \mathbf{J} - \mathbf{L}_{\rm elec}$, and the electron sector is the clean scalar H_e (Proof 2).

Watson reciprocal inertia

Invert (3.19) for $\boldsymbol{\omega}$: write $\mathbf{L}_N^{\mathrm{vib}} = \boldsymbol{\pi} - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}}\boldsymbol{\omega}$ with

$$\boldsymbol{\pi} = \sum_b \boldsymbol{\sigma}_b P_b \;\; \text{(field-free vibrational ang. mom.)}, \qquad \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}} = \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^{\mathsf{T}}. \tag{3.20a}$$

Then $\mathbf{J} - \mathbf{L}_{\mathrm{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})\boldsymbol{\omega} + \boldsymbol{\pi}$, so

$$\boxed{\boldsymbol{\omega} = \boldsymbol{\mu}\big(\mathbf{J} - \mathbf{L}_{\mathrm{elec}} - \boldsymbol{\pi}\big), \qquad \boldsymbol{\mu} \equiv (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}.} \tag{3.20}$$

$\boldsymbol{\mu}$ is the Watson reciprocal-inertia tensor — built from *nuclear* quantities only, with no Wilson G-matrix.

The exact classical molecular Hamiltonian

Legendre back. The rovibrational block gives $\frac{1}{2}(\mathbf{J}-\mathbf{L}_{\mathrm{elec}}-\boldsymbol{\pi})^{\mathsf{T}}\boldsymbol{\mu}(\mathbf{J}-\mathbf{L}_{\mathrm{elec}}-\boldsymbol{\pi})+\frac{1}{2}\sum_b P_b^2$; the electron sector gives H_e (Proof 2):

$$\boxed{\begin{aligned} H_{\mathrm{int}} &= \frac{1}{2}\big(\mathbf{J}-\mathbf{L}_{\mathrm{elec}}-\boldsymbol{\pi}\big)^{\mathsf{T}}\boldsymbol{\mu}(Q)\big(\mathbf{J}-\mathbf{L}_{\mathrm{elec}}-\boldsymbol{\pi}\big)+\frac{1}{2}\sum_b P_b^2 \\ &\quad+\sum_i \frac{\bar{\mathbf{p}}_i^{\,2}}{2m_e}+\frac{1}{2M_N}\Big(\sum_i \bar{\mathbf{p}}_i\Big)^2+V_{NN}+V_{Ne}+V_{ee}. \end{aligned}} \tag{3.22}$$

Exact, classical, in BF + Eckart coordinates. Now we quantize.

The quantization problem

Primer: wavefunctions live in a space, L^2

In quantum mechanics a state is a wavefunction $\psi(q)$ — and wavefunctions add and scale like vectors, so they form a vector space. To measure "length" and "angle" we need an inner product; for two states φ, ψ,

$$\langle \varphi | \psi \rangle \equiv \int \varphi^*(q)\, \psi(q)\, dq \qquad \text{(Dirac ``bra–ket'' notation).}$$

The squared length is the norm, and a physical state is normalized:

$$\|\psi\|^2 = \langle \psi | \psi \rangle = \int |\psi(q)|^2\, dq = 1 \qquad \text{(total probability} = 1\text{).}$$

What is L^2?

L^2 is the space of square-integrable functions — those with finite $\int |\psi|^2 < \infty$, so the norm makes sense. Operators like $\hat{H}$ act on this space, and observables must be Hermitian, $\hat{O}^\dagger = \hat{O}$, to have real eigenvalues.

Primer: the measure — what you integrate against

The inner product needs a rule for "how much volume" each point carries — the measure. In plain Cartesian coordinates it is $dq = dx\,dy\,dz$, but after a change of variables it picks up the Jacobian:

$$\int f\,dx\,dy\,dz = \int f\,\underbrace{r^2\sin\theta}_{\text{Jacobian}}\,dr\,d\theta\,d\phi \quad\Longrightarrow\quad \langle\varphi|\psi\rangle = \int \varphi^*\psi\,\underbrace{w(q)}_{\text{measure}}\,d^nq.$$

Normalization is always against this measure: $\int |\psi|^2\,w\,d^nq = 1$. We write the space as $L^2(w\,d^nq)$.

Why it matters here

Hermiticity depends on the measure. On a curved measure $-i\hbar\partial$ is *not* Hermitian (next slides show this for $w = r^2\sin\theta$). Getting the operator right on the correct measure is exactly the quantization problem — and its cure is the Podolsky rule. For the molecule the measure will be $\sqrt{g}\,d^nq$ (later reshaped to the flat $d\mu_W$).

A worked example: one particle in spherical coordinates

Take a single free particle of mass m. Cartesian: $\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$. In spherical coordinates (r, θ, ϕ),

$$\mathcal{L} = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right).$$

Canonical momenta $p_a = \partial\mathcal{L}/\partial\dot{q}^a$:

$$p_r = m\dot{r}, \qquad p_\theta = mr^2\dot{\theta}, \qquad p_\phi = mr^2\sin^2\theta\,\dot{\phi}.$$

Legendre transform $\Rightarrow$ the classical Hamiltonian

$$\boxed{H = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + \frac{p_\phi^2}{2mr^2\sin^2\theta}.}$$

The coordinates are *curvilinear*: p_θ, p_ϕ are not mass$\times$velocity, and the coefficients depend on r, θ.

Naive quantization gives the wrong operator

Substitute $p_a \to -i\hbar\,\partial_a$ into H:

$$\hat{H}_{\text{naive}} = -\frac{\hbar^2}{2m}\left[\partial_r^2 + \frac{1}{r^2}\partial_\theta^2 + \frac{1}{r^2\sin^2\theta}\partial_\phi^2\right].$$

But the true kinetic operator is $-\frac{\hbar^2}{2m}\nabla^2$, with the familiar spherical Laplacian

$$\nabla^2 = \underbrace{\frac{1}{r^2}\partial_r(r^2\partial_r)}_{\partial_r^2+\frac{2}{r}\partial_r} + \underbrace{\frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta)}_{\frac{1}{r^2}(\partial_\theta^2+\cot\theta\,\partial_\theta)} + \frac{1}{r^2\sin^2\theta}\partial_\phi^2.$$

Naive quantization is missing the first-derivative terms $\frac{2}{r}\partial_r$ and $\frac{\cot\theta}{r^2}\partial_\theta$. Where did they go?

Why it fails: $-i\hbar\partial$ is not Hermitian on the curved measure

The physical inner product carries the volume Jacobian:

$$\langle\varphi|\psi\rangle = \int \varphi^*\psi \underbrace{r^2\sin\theta}_{\text{Jacobian}} dr\, d\theta\, d\phi.$$

Is $\hat{p}_r = -i\hbar\partial_r$ Hermitian here? Integrate by parts in r (boundary terms vanish):

$$\int \varphi^*(-i\hbar\partial_r\psi)\, r^2\, dr = \int(-i\hbar\partial_r\varphi)^*\psi\, r^2\, dr - i\hbar\int \varphi^*\psi \underbrace{(\partial_r r^2)}_{=(2/r)\, r^2} dr,$$

so

$$\boxed{\hat{p}_r^\dagger = -i\hbar\Big(\partial_r + \frac{2}{r}\Big) \neq \hat{p}_r.}$$

Not Hermitian $\Rightarrow$ $\hat{p}_r^2$ built this way isn't either $\Rightarrow$ no guaranteed real spectrum. The naive recipe ignored the measure. (Same for θ: the $\cot\theta$ term.)

The Podolsky / Laplace–Beltrami rule

Cure: read the kinetic energy as a quadratic form, $T = \frac{1}{2}g_{ab}(q)\dot{q}^a\dot{q}^b = \frac{1}{2}g^{ab}(q)\, p_a p_b$, which *defines* the metric g_{ab} and the measure $\sqrt{g}\, d^n q$. The operator Hermitian on $L^2(\sqrt{g}\, d^n q)$ *by construction* is

$$\boxed{\hat{T} = -\frac{\hbar^2}{2}\frac{1}{\sqrt{g}}\,\partial_a\big(\sqrt{g}\, g^{ab}\,\partial_b\big), \qquad g = \det g_{ab}.} \tag{4.5}$$

One integration by parts gives the manifestly symmetric, positive form

$$\int \varphi^* \,\hat{T}\psi\,\sqrt{g}\, d^n q = \frac{\hbar^2}{2}\int (\partial_a\varphi)^* g^{ab}(\partial_b\psi)\,\sqrt{g}\, d^n q. \tag{4.6}$$

The $\sqrt{g}$ factors are exactly the measure the naive recipe forgot; for flat Cartesian $g_{ab} = m\delta_{ab}$ it is $-\frac{\hbar^2}{2m}\nabla^2$.

Podolsky recovers the correct Hamiltonian

For the spherical particle, $\mathcal{L}$ gives $g_{ab} = \mathrm{diag}(m,\ mr^2,\ mr^2 \sin^2\theta)$, so

$$\sqrt{g} = m^{3/2} r^2 \sin\theta, \qquad g^{ab} = \mathrm{diag}\left(\tfrac{1}{m},\ \tfrac{1}{mr^2},\ \tfrac{1}{mr^2\sin^2\theta}\right).$$

Insert into (4.5); the constant $m^{3/2}$ cancels between $\sqrt{g}$ and $1/\sqrt{g}$:

$$\hat{T} = -\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\partial_r(r^2\partial_r) + \frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \frac{1}{r^2\sin^2\theta}\partial_\phi^2\right] = -\frac{\hbar^2}{2m}\nabla^2. \checkmark$$

The $\frac{2}{r}\partial_r$ and $\cot\theta\,\partial_\theta$ are restored automatically — by the measure, not by hand.

The molecule is the same story

Curvilinear $\{q^A\} = \{\phi, \theta, \chi, Q_1, \ldots, Q_f\}$, $T = \frac{1}{2}g^{AB}p_A p_B$, curved measure $\sqrt{g}\,d^n q$. We now build the three objects — metric g_{AB}, determinant g, inverse g^{AB} — explicitly, then apply (4.5). *Next section.*

Recipe: Podolsky / Laplace–Beltrami quantization

General recipe — curvilinear $\{q^i\}$:

1. **Metric** g_{ij}: read off $T = \frac{1}{2} g_{ij} \dot{q}^i \dot{q}^j$, or $g_{ij} = \sum_a m_a (\partial \mathbf{r}_a / \partial q^i) \cdot (\partial \mathbf{r}_a / \partial q^j)$.
2. **Determinant & measure**: $g = \det g_{ij} \Rightarrow$ Hilbert space $L^2(\sqrt{g}\, d^n q)$.
3. **Inverse** $g^{ij} = (g_{ij})^{-1}$.
4. **Podolsky operator** (Hermitian on $L^2(\sqrt{g}\, d^n q)$ by construction):

$$\hat{T} = -\frac{\hbar^2}{2} \frac{1}{\sqrt{g}} \partial_i \left(\sqrt{g}\, g^{ij} \partial_j \right);$$

   V and coordinate functions → multiplication.
5. **(Optional) rescale** $\psi \to g^{1/4} \psi$ to a flatter measure; leftover c-number = pseudopotential.

Instance: free particle in (r, θ, ϕ)

1. $g_{ij} = \mathrm{diag}(m,\ mr^2,\ mr^2 \sin^2 \theta)$
2. $\sqrt{g} = m^{3/2} r^2 \sin \theta$
3. g^{ij}: reciprocals of g_{ii}
4. $\hat{T} = -\dfrac{\hbar^2}{2m} \nabla^2$ ✓
5. flat ⇒ no pseudopotential

Molecule (next): metric (4.15) → measure (4.21) → inverse (4.22) → Podolsky → Watson $\hat{U}$.

The ro-vibrational geometry

Euler-angle kinematics, explicitly

zyz convention $\mathbf{S} = R_z(\phi)R_y(\theta)R_z(\chi)$ (4.7). Differentiating $\mathbf{S}$ gives $\boldsymbol{\omega} = \mathsf{E}(\Omega)\dot{\boldsymbol{\Omega}}$ and, solving the linear system, $\dot{\boldsymbol{\Omega}} = \mathsf{E}^{-1}\boldsymbol{\omega}$:

$$\mathsf{E} = \begin{pmatrix} -\sin\theta\cos\chi & \sin\chi & 0 \\ \sin\theta\sin\chi & \cos\chi & 0 \\ \cos\theta & 0 & 1 \end{pmatrix}, \qquad \mathsf{E}^{-1} = \begin{pmatrix} -\frac{\cos\chi}{\sin\theta} & \frac{\sin\chi}{\sin\theta} & 0 \\ \sin\chi & \cos\chi & 0 \\ \frac{\cos\theta\cos\chi}{\sin\theta} & -\frac{\cos\theta\sin\chi}{\sin\theta} & 1 \end{pmatrix} \tag{4.10, 4.12}$$

with

$$\det\mathsf{E} = -\sin\theta. \tag{4.11}$$

The $1/\sin\theta$ entries are the Euler coordinate singularity at the poles; they are compensated by $|\det\mathsf{E}| = \sin\theta$ in the volume element. (4.10)–(4.12) are the only Euler-angle facts used from here on.

The ro-vibrational metric tensor

The classical BF kinetic energy $2T_{\mathrm{rv}} = \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + 2\boldsymbol{\omega}^T \sum_b \boldsymbol{\sigma}_b \dot{Q}_b + \sum_b \dot{Q}_b^2$ (4.13), re-expressed in the *coordinate* velocities via $\boldsymbol{\omega} = \mathsf{E}\dot{\boldsymbol{\Omega}}$, defines the metric:

$$\boxed{g_{AB} = \begin{pmatrix} \mathsf{E}^T \mathbf{I}_N \mathsf{E} & \mathsf{E}^T \boldsymbol{\sigma} \\ \boldsymbol{\sigma}^T \mathsf{E} & \mathbb{1}_f \end{pmatrix}} \tag{4.15}$$

- Euler–Euler block: inertia pulled back through the kinematic matrix.
- Euler–mode block: Coriolis coupling $\boldsymbol{\sigma}_b = \sum_a \zeta_{ab} Q_a$ (constant ζ's).
- mode–mode block: *identity* — the gift of mass-weighted rectilinear normal coordinates.

Every entry is an explicit function of (θ, χ, Q).

Electrons: flat, constant, decoupled from the measure

Including electron velocities, the full mass matrix has a *constant* electron block D. Eliminating it (Schur complement = completing the square) removes all ω-dependence:

$$\mathsf{C}\mathsf{D}^{-1}\mathsf{C}^{\mathsf{T}} = \mathbf{I}_e - \mathbf{I}_X \implies \det \mathsf{M} = \underbrace{\det \mathsf{D}}_{\text{const}} \cdot \det \begin{pmatrix} \mathbf{I}_N & \boldsymbol{\sigma} \\ \boldsymbol{\sigma}^{\mathsf{T}} & \mathbb{1}_f \end{pmatrix} \tag{4.17}$$

— the same $\mathbf{I}_e/\mathbf{I}_X$ cancellation as the classical cross-check of § 3.6.

- Electron measure is the flat $\prod_i d^3\bar{\mathbf{r}}_i$; $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ is Hermitian on it.
- $\hat{L}_{\mathrm{elec},\alpha}$ is Hermitian *without symmetrization*: its vector field is divergence-free, $\sum_{i\gamma} \partial_{\bar{r}_{i\gamma}}(\epsilon_{\alpha\beta\gamma}\bar{r}_{i\beta}) = 0$ (4.18).

The entire quantization problem lives in the ro-vibrational sector; electrons enter only via the flat kinetic operator and the shift $\hat{J}_\alpha \to \hat{J}_\alpha - \hat{L}_{\mathrm{elec},\alpha}$.

Determinant and volume element

Block row reduction gives $\det\begin{pmatrix} \mathsf{a} & \mathsf{b} \\ \mathsf{b}^T & \mathbb{1} \end{pmatrix} = \det(\mathsf{a} - \mathsf{b}\mathsf{b}^T)$ (4.19). Applied to (4.15):

$$\boxed{\det g_{AB} = \sin^2\theta\, \gamma(Q), \qquad \gamma \equiv \det \mathbf{I}', \qquad \mathbf{I}' \equiv \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T.} \tag{4.20}$$

The Watson-modified inertia appears by itself

The determinant is governed by $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T$ — *not* the bare $\mathbf{I}_N$: the Coriolis coupling reduces the rotational determinant. It is the same $\mathbf{I}'$ whose inverse $\boldsymbol{\mu}$ is the classical rotational kernel (3.20).

$$\sqrt{g}\, d^{3+f} q = \underbrace{\sin\theta\, d\phi\, d\theta\, d\chi}_{\text{Haar measure on } SO(3)} \cdot \underbrace{\sqrt{\gamma(Q)}\, \textstyle\prod_b dQ_b}_{\text{rot.–vib. Jacobian}} \cdot \underbrace{\textstyle\prod_i d^3\bar{\mathbf{r}}_i}_{\text{flat}} \tag{4.21}$$

The inverse metric

Claim, then *verify* by block multiplication using only $\boldsymbol{\mu}\mathbf{I}' = \mathbb{1}$ [(4.23)]:

$$\boxed{g^{AB} = \begin{pmatrix} \mathsf{E}^{-1}\boldsymbol{\mu}\,\mathsf{E}^{-T} & -\mathsf{E}^{-1}\boldsymbol{\mu}\,\boldsymbol{\sigma} \\ -\boldsymbol{\sigma}^{T}\boldsymbol{\mu}\,\mathsf{E}^{-T} & \mathbb{1}_f + \boldsymbol{\sigma}^{T}\boldsymbol{\mu}\,\boldsymbol{\sigma} \end{pmatrix}, \qquad \boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{T})^{-1}.} \tag{4.22}$$

Two inversions, one computation

The Watson reciprocal inertia $\boldsymbol{\mu}$ — introduced classically by inverting the momentum–velocity relation (3.20) — reappears as the geometric inverse of the metric.

e.g. top-left: $\mathsf{E}^{T}\mathbf{I}_N\mathsf{E}\cdot\mathsf{E}^{-1}\boldsymbol{\mu}\mathsf{E}^{-T} - \mathsf{E}^{T}\boldsymbol{\sigma}\cdot\boldsymbol{\sigma}^{T}\boldsymbol{\mu}\mathsf{E}^{-T} = \mathsf{E}^{T}(\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{T})\boldsymbol{\mu}\mathsf{E}^{-T} = \mathbb{1}_3$. ✓

Quantizing rotation & Coriolis coupling

Why the $\hat{J}_\alpha$ look unlike the usual angular momentum

The rotational operators below differ from the $\hat{L}$ of introductory QM for *three* reasons:

1. **Three angles, not two.** A rigid body needs (ϕ, θ, χ), so besides $\partial_\phi, \partial_\theta$ there is a third generator $\hat{J}_3 = -i\hbar\partial_\chi$, and the three mix.
2. **Body-fixed, not space-fixed.** They are components along the *molecule's* axes, $\hat{J}_\alpha^{\mathrm{BF}} = S_{\beta\alpha}(\Omega)\hat{J}_\beta^{\mathrm{SF}}$; since **S** depends on the angles, their commutator carries the anomalous sign $[\hat{J}_\alpha, \hat{J}_\beta] = -i\hbar\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma$ — which we *never use*.
3. **Convention-dependent form.** The explicit trig coefficients are fixed by our *zyz* choice $\mathbf{S} = R_z(\phi)R_y(\theta)R_z(\chi)$; another convention relabels them (*zyz* is itself standard — Wigner, Zare).

Nothing is postulated — the operators fall out of the Laplace–Beltrami expansion next.

Laplace–Beltrami on the rotational block ⇒ $\hat{J}_\alpha$

Put the measure $\sqrt{g} = \sin\theta\sqrt{\gamma}$ and the rotational inverse-metric block $g^{\Omega\Omega}_{st} = (\mathsf{E}^{-1})_{s\alpha}\mu_{\alpha\beta}(\mathsf{E}^{-1})_{t\beta}$ into the Podolsky/LB operator (4.5). Since μ, $\sqrt{\gamma}$ depend on Q only, they pass through ∂_Ω:

$$\hat{T}_{\mathrm{rot}} = -\frac{\hbar^2}{2}\,\mu_{\alpha\beta}\,\frac{1}{\sin\theta}\,\partial_s\!\big(\sin\theta\,(\mathsf{E}^{-1})_{s\alpha}\,(\mathsf{E}^{-1})_{t\beta}\,\partial_t\big).$$

The combination $(\mathsf{E}^{-1})_{s\alpha}\partial_{\Omega_s}$ appears twice — name it:

$$\boxed{\hat{J}_\alpha \equiv -i\hbar\sum_s(\mathsf{E}^{-1})_{s\alpha}\partial_{\Omega_s}:\quad \begin{aligned} \hat{J}_1 &= -i\hbar[-\tfrac{\cos\chi}{\sin\theta}\partial_\phi + \sin\chi\,\partial_\theta + \tfrac{\cos\theta\cos\chi}{\sin\theta}\partial_\chi],\\ \hat{J}_2 &= -i\hbar[\ \tfrac{\sin\chi}{\sin\theta}\partial_\phi + \cos\chi\,\partial_\theta - \tfrac{\cos\theta\sin\chi}{\sin\theta}\partial_\chi],\\ \hat{J}_3 &= -i\hbar\,\partial_\chi,\end{aligned}} \tag{4.24}$$

read row-by-row off E^{-1} (4.12). One leftover term $\partial_s(\sin\theta\,\mathsf{E}^{-1})$ remains — next.

The divergence identity makes it clean

Expanding the outer derivative gives the clean double-derivative *plus* a first-order remainder:

$$\hat{T}_{\text{rot}} = \tfrac{1}{2}\mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta - \tfrac{\hbar^2}{2}\mu_{\alpha\beta}\underbrace{\left[\tfrac{1}{\sin\theta}\partial_s\big(\sin\theta(\mathsf{E}^{-1})_{s\alpha}\big)\right]}_{\text{divergence (4.26)}}(\mathsf{E}^{-1})_{t\beta}\partial_t.$$

The divergence vanishes term by term:

$$\alpha{=}1:\ \partial_\phi(-\cos\chi) + \partial_\theta(\sin\theta\sin\chi) + \partial_\chi(\cos\theta\cos\chi) = 0,$$

$$\alpha{=}2:\ \partial_\phi(\sin\chi) + \partial_\theta(\sin\theta\cos\chi) + \partial_\chi(-\cos\theta\sin\chi) = 0, \quad \alpha{=}3:\ \partial_\chi\sin\theta = 0.$$

Why the rotation block quantizes so cleanly (for the advanced reader)

Podolsky's rule *should* spawn $\partial\sqrt{g}$ and $\partial g^{\Omega\Omega}$ first-order terms — but they collapse into this one divergence, zero because the rotation generators are divergence-free on $\sin\theta\, d\Omega$. Hence $\hat{T}_{\text{rot}} = \tfrac{1}{2}\mu_{\alpha\beta}\hat{J}_\alpha\hat{J}_\beta$, Hermitian, no pseudopotential (4.29).

The Coriolis (rotation–vibration) block — also clean

The off-diagonal blocks $g^{\Omega Q} = -\mathsf{E}^{-1}\boldsymbol{\mu}\boldsymbol{\sigma}$ give the cross term (4.30):

$$\hat{T}_{\mathrm{Cor}} = -\tfrac{1}{2}\Big[\hat{J}_\alpha\, \mu_{\alpha\beta}\, \hat{\pi}_\beta + \tfrac{1}{\sqrt{\gamma}}\, \hat{P}_b\, \sqrt{\gamma}\, \sigma_{\alpha b}\mu_{\alpha\beta}\, \hat{J}_\beta\Big], \qquad \hat{\pi}_\alpha = \textstyle\sum_b \sigma_{\alpha b}(Q)\hat{P}_b.$$

The two groups are adjoints of each other on the measure (4.21) — the $\gamma^{\pm 1/2}$ are exactly what integration by parts over $\sqrt{\gamma}\, dQ$ produces — so their sum is Hermitian *automatically*, with no residue.

Section result

Rotation (4.29) and Coriolis (4.30) quantize with **no** pseudopotential — the divergence identity and the adjoint-pair structure do all the work. The flat-measure rescaling of the next section leaves both untouched (giving the symmetric $-\tfrac{1}{2}(\hat{J}\mu\hat{\pi} + \hat{\pi}\mu\hat{J})$, 4.35). Every pseudopotential comes from the vibrational block.

The vibrational block & the Watson pseudopotential

The vibrational block is different

The vibrational inverse-metric block is $g^{QQ} = \mathbb{1} + \boldsymbol{\sigma}^{\mathsf{T}}\boldsymbol{\mu}\boldsymbol{\sigma}$; the LB operator on the curved measure $\sqrt{\gamma}$ is

$$\hat{T}_{\mathrm{vib}} = -\frac{\hbar^2}{2}\,\frac{1}{\sqrt{\gamma}}\,\partial_{Q_b}\big(\sqrt{\gamma}\,g^{QQ}_{bc}\,\partial_{Q_c}\big).$$

Expand the δ_{bc} part by one product rule:

$$-\tfrac{\hbar^2}{2}\tfrac{1}{\sqrt{\gamma}}\partial_{Q_b}\big(\sqrt{\gamma}\,\partial_{Q_b}\psi\big) = \tfrac{1}{2}\hat{P}_b^2\psi \underbrace{-\tfrac{\hbar^2}{2}\big(\tfrac{1}{2}\partial_{Q_b}\ln\gamma\big)\partial_{Q_b}\psi}_{\text{first-order remainder}}.$$

Unlike rotation, this remainder does not vanish: $\partial_{Q_b}\ln\gamma \neq 0$ because $\gamma = \det \mathbf{I}'(Q)$ genuinely depends on the shape, and the bare $\hat{P}_b$ is *not* Hermitian on $\sqrt{\gamma}\,dQ$. How to handle it?

Warm-up: the radial part of the spherical Laplacian

Read the radial coordinate as an effective one-body *vibration* (a diatomic bond). Its Laplace–Beltrami operator, on the curved measure $r^2\,dr$, is

$$\hat{T}_r = -\frac{\hbar^2}{2m}\frac{1}{r^2}\partial_r(r^2\partial_r), \qquad \int |R|^2\, r^2\, dr = 1.$$

But vibrational wavefunctions are conventionally normalized on a *flat* measure. Rescale $u = rR$ (i.e. $\psi \to g^{1/4}\psi$, $\sqrt{g} \propto r^2$) so $\int |u|^2\, dr = 1$:

$$\boxed{\hat{T}_r \xrightarrow{u=rR} -\frac{\hbar^2}{2m}\frac{d^2}{dr^2} \quad \text{on the flat measure } dr.}$$

The awkward $\frac{2}{r}\partial_r$ is gone — absorbed into the rescaling. *This* move is what, in general, leaves a pseudopotential behind.

Two schemes: Laplace–Beltrami vs Podolsky

The *same* operator, in two representations:

Laplace–Beltrami

wavefn on curved $\sqrt{g}\, d^n q$

$-\frac{\hbar^2}{2\sqrt{g}}\partial(\sqrt{g}\, g^{ij}\partial)$

no explicit potential

Podolsky (flat measure)

$\psi_W = g^{1/4}\psi$ on flat $d^n q$

$\frac{1}{2}\hat{p}_i g^{ij}\hat{p}_j + U$

extra c-number $U =$ pseudopotential

Related by conjugation with $g^{1/4}$; U is the bookkeeping cost of the flat measure.

Radial case: $U = 0$

For $\gamma \propto r^2$ the residual cancels ($u = rR$ is exact) — like the rigid rotor. A genuine shape-dependent $\gamma(Q)$ does not cancel: that is where the Watson potential is born.

Rescaling the molecular vibrational block

Same move with $\gamma = \det \mathbf{I}'(Q)$: absorb the shape factor, $\psi_W = \gamma^{1/4}\psi$, so the vibrational measure is flat, $\prod_b dQ_b$ (normal-mode space treated as flat). Conjugating $\hat{T}_{\mathrm{vib}}$ (one product rule per line):

$$\begin{aligned} \partial_{Q_b}(\gamma^{-1/4}\psi_W) &= \gamma^{-1/4}[\partial_{Q_b}\psi_W - \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)\psi_W], \\ \partial_{Q_b}(\gamma^{1/4}[\cdots]) &= \gamma^{1/4}[\partial^2_{Q_b}\psi_W - \tfrac{1}{16}(\partial_{Q_b}\ln\gamma)^2\psi_W - \tfrac{1}{4}(\partial^2_{Q_b}\ln\gamma)\psi_W]. \end{aligned}$$

The cross terms cancel; the bare $\frac{1}{2}\sum_b \hat{P}_b^2$ survives *plus* a multiplicative residue

$$U_{\mathrm{vib}} = \frac{\hbar^2}{8}\sum_b \left[\partial^2_{Q_b}\ln\gamma + \tfrac{1}{4}(\partial_{Q_b}\ln\gamma)^2\right] \quad (+\,U_{\mathrm{Cor}} \text{ from the } \boldsymbol{\sigma}^{\mathsf{T}}\boldsymbol{\mu}\boldsymbol{\sigma} \text{ dressing}).$$

Now the residue is nonzero — the promised pseudopotential.

Collapse to the Watson trace

The residue $U_{\mathrm{vib}} + U_{\mathrm{Cor}}$ collapses through two exact Q-calculus identities. **Jacobi:** $\partial_{Q_b} \ln \gamma = \mathrm{tr}(\boldsymbol{\mu}\, \partial_{Q_b} \mathbf{I}')$. **Watson's closed form (4.37):** $\mathbf{I}' = A(\mathbf{I}^e)^{-1} A$ with $A = \mathbf{I}^e + \frac{1}{2} \sum_b a^b Q_b$, so every derivative reduces to constants; the Eckart sum rules fix the remainder:

$$\boxed{\hat{U}(Q) = -\frac{\hbar^2}{8} \sum_{\alpha=1}^{3} \mu_{\alpha\alpha}(Q).} \tag{4.38}$$

- A pure c-number from the *measure* — kernel ordering never makes one.
- Electrons don't modify it ($\gamma, \boldsymbol{\mu}$ depend on Q alone).
- Special to *rectilinear* normal coordinates (Watson's closed form).

The complete quantum Hamiltonian

Electronic kinetic operator

In BF Cartesian electron coordinates $\hat{\bar{\mathbf{p}}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ is already Hermitian on the flat measure, so the electron + mass-polarization piece quantizes directly:

$$\hat{T}_e = \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2 = \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^2}{2\mu_e} + \frac{1}{M_N}\sum_{i<l} \hat{\bar{\mathbf{p}}}_i \cdot \hat{\bar{\mathbf{p}}}_l. \qquad (4.39,\ 4.40)$$

The diagonal part renormalizes the electron mass, $\frac{1}{m_e} + \frac{1}{M_N} = \frac{1}{\mu_e}$; the off-diagonal two-electron term is the mass polarization *sensu stricto* (nuclear recoil).

The exact non-relativistic molecular Hamiltonian

Collecting the flat-measure blocks — rotational (4.29), Coriolis (4.35), vibrational, pseudopotential (4.38), electronic (4.39), and the shift $\hat{J}_\alpha \to \hat{J}_\alpha - \hat{L}_{\mathrm{elec},\alpha}$:

$$\boxed{\begin{aligned}
\hat{H}_{\mathrm{int}} = {} & \tfrac{1}{2}\sum_{\alpha\beta}\big(\hat{J}_\alpha - \hat{L}_{\mathrm{elec},\alpha} - \hat{\pi}_\alpha\big)\mu_{\alpha\beta}(Q)\big(\hat{J}_\beta - \hat{L}_{\mathrm{elec},\beta} - \hat{\pi}_\beta\big) && \text{rotation–vibration–electron} \\
& + \tfrac{1}{2}\sum_b \hat{P}_b^2 && \text{vibration} \\
& + \sum_i \frac{\hat{\bar{\mathbf{p}}}_i^{\,2}}{2m_e} + \frac{1}{2M_N}\Big(\sum_i \hat{\bar{\mathbf{p}}}_i\Big)^2 && \text{electrons + mass pol.} \\
& - \frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}(Q) && \text{Watson pseudopotential} \\
& + V_{NN} + V_{Ne} + V_{ee} && \text{Coulomb}
\end{aligned}} \tag{4.41}$$

on $L^2(d\mu_W)$, $d\mu_W = \sin\theta\, d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\bar{\mathbf{r}}_i$. Every operator is an explicit differential operator, Hermitian by integration by parts.

Remark: the anomalous commutator — not needed

The traditional route to (4.41) (Klein 1929; Van Vleck 1951) derives the *anomalous* algebra of the body-fixed angular momentum,

$$[\hat{J}_\alpha, \hat{J}_\beta] = -\,i\hbar\,\epsilon_{\alpha\beta\gamma}\,\hat{J}_\gamma \qquad \text{(note the } - \text{ sign)},$$

and uses it to control the operator ordering of the rotational kernel.

Nothing here used it

- $\hat{J}_\alpha$ entered only as the *shorthand* (4.24) for explicit Euler-angle derivatives.
- Hermiticity came from integration by parts + the divergence identity (4.26).
- The pseudopotential came from the Q-derivatives of the volume element.

The anomalous algebra is a correct property of the operators (4.24) — but it is a consequence of the coordinate representation, not an input to the quantization.

Born–Huang expansion and the BO truncation

At fixed Q, the clamped-nuclei electronic problem $\hat{H}_{\mathrm{el}}(Q)|\phi_n(Q)\rangle = E_n^{\mathrm{el}}(Q)|\phi_n(Q)\rangle$ (4.42–4.43) provides a complete parametric basis. The exact molecular wavefunction expands as

$$|\Psi\rangle = \sum_n |\chi_n\rangle \otimes |\phi_n(Q)\rangle \tag{4.45}$$

— not an ansatz: it follows from completeness at each Q. Projecting the Schrödinger equation gives coupled channels (4.46) with non-adiabatic couplings $\hat{\Lambda}_{nm}$ (4.47).

Born–Oppenheimer = diagonal truncation

Set $\hat{\Lambda}_{nm} \to 0$: single-surface nuclear dynamics (4.48) on the potential energy surface

$$V_n^{\mathrm{BO}}(Q) = E_n^{\mathrm{el}}(Q) + V_{NN}(Q) + \hat{U}(Q). \tag{4.49}$$

A controlled approximation *of an exact operator* — breaking down at conical intersections.

Summary

The journey, in four stages

1. Classical: lab H (1.1) $\rightarrow$ one mixed mass-centered point transformation $\rightarrow$ H (2.12): COM exactly split for any $\mathbf{P}_{\mathrm{cm}}$, mass polarization included — no boost, no rest frame.
2. Body-Fixed: velocity-form Lagrangian $(\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X)$ $\rightarrow$ key identity $\mathbf{J} - \mathbf{L}_{\mathrm{elec}} = \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\mathrm{vib}}$ $(\mathbf{I}_e, \mathbf{I}_X$ cancel) $\rightarrow$ classical H_{int} (3.22) with Watson $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathsf{T}})^{-1}$.
3. Quantize, coordinate-natively: metric (4.15) $\rightarrow$ measure (4.21) $\rightarrow$ inverse (4.22) $\rightarrow$ Podolsky blocks $\rightarrow$ $\gamma^{1/4}$ rescaling $\rightarrow$ Watson $\hat{U} = -\frac{\hbar^2}{8}\sum_\alpha \mu_{\alpha\alpha}$ (4.38). Hermiticity by integration by parts + divergence identity (4.26) — no angular-momentum algebra.
4. Quantum $\hat{H}$: the exact non-relativistic $\hat{H}_{\mathrm{int}}$ (4.41).

What we did — and did not — assume

$\hat{H}_{\mathrm{int}}$ (4.41) is exact. The only inputs were:

- non-relativistic dynamics;
- the Eckart frame (a convention);
- $3N_{\mathrm{nucl}} - 6$ normal coordinates (an exact parametrization).

No Born–Oppenheimer, no clamped nuclei, no neglect of mass polarization, no harmonic assumption — and no anomalous commutator as input.

Outlook

(4.41) feeds the Born–Huang expansion (4.45), whose diagonal truncation recovers the familiar single-surface picture (4.48)–(4.49) — a controlled approximation *of an exact operator*.

The full molecular Hamiltonian:
exact, quantized coordinate-natively,
and ready to approximate.